PRESENCIFY: A CPU Only Face Recognition Attendance Prototype with Fixed Confidence Thresholds and an Uncertainty Review Band

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Abstract: *PRESENCIFY evaluates an edge-based facial recognition attendance prototype engineered to execute entirely on commodity laptop CPUs without dedicated graphics acceleration. The system couples MediaPipe BlazeFace for face localization with an EdgeFace XS backbone, producing 512-dimensional L2-normalized feature vectors matched against enrolled templates via cosine similarity, persisted in SQLite and administered through a local Streamlit dashboard. Rather than presuming automated infallibility, this study audits the operational tension between algorithmic error rates and manual instructor verification workloads under fixed decision boundaries. In a controlled benchmark of 166 static probe images, an immediate acceptance cutoff of 0.65 correctly verifies 65 genuine probes but misclassifies 7 genuine probes as peer identities. Crucially, this single cutoff admits 18 unauthorized impostor probes into the register, yielding a 27.27% false acceptance rate. Introducing an intermediate uncertainty band ([0.52, 0.65)) intercepts 33 borderline impostors, diverting 33.73% of total verification traffic (56 probes) to manual review. Counterfactual analysis using ground-truth labels indicates that an error-free human reviewer could restore 12 genuine student records, raising cumulative accuracy to 75.30%. The architecture explicitly lacks presentation attack detection (PAD_UNAVAILABLE), and initial subject data incorporates benchmark image fallbacks. These empirical findings quantify realistic deployment trade-offs between automated throughput and teacher oversight on edge hardware.*

Index Terms: face recognition, attendance system, edge computing, confidence thresholds, human review, presentation attack detection.

I. INTRODUCTION

Manual roll-call recording during university lectures consumes instructional time and remains vulnerable to peer proxy signing. Automating verification via computer vision has gained interest in campus settings [1]–[3], yet practical deployment in collegiate environments encounters severe hardware constraints. Departmental lecture kiosks and faculty workstations commonly rely on basic dual-core or quad-core mobile processors without discrete GPUs. Under these computational boundaries, practical attendance kiosks must deploy lightweight neural architectures that sustain interactive frame rates locally while ensuring sensitive student facial embeddings remain contained within edge devices.

Facial feature extractors map aligned face crops into metric spaces where cosine similarity yields an uncalibrated scalar distance rather than a posterior probability of identity. Establishing a single static decision threshold therefore imposes conflicting operational demands. Elevating the cutoff suppresses unauthorized entries but triggers frequent false rejections of legitimate attendees; conversely, lowering the boundary limits student inconvenience while permitting unauthorized peers to register attendance undetected. Routing borderline similarity scores to an uncertainty review queue provides an operational buffer, yet this triage mechanism directly transfers administrative inspection overhead onto teaching staff. Determining whether static confidence boundaries and human review workflows can maintain defensible classroom security without creating unsustainable teacher workloads remains an open empirical challenge.

We implement and evaluate PRESENCIFY to analyze this operational balance under controlled edge conditions. The contributions of this work are threefold: delivering an integrated offline attendance kiosk tailored for x86 CPU processors without external runtime dependencies, quantifying classification errors alongside instructor review burdens across distinct similarity intervals, and isolating automated pipeline accuracy from counterfactual performance gains modeled through oracle-simulated review.

II. LITERATURE REVIEW

A. Lightweight recognition and detection

Recent research in edge biometrics emphasizes model compression techniques that facilitate real-time facial feature extraction on resource-limited hardware [4]–[6]. George et al. designed EdgeFace, a hybrid convolutional-transformer architecture that compresses deep biometric representations into compact checkpoints suitable for mobile deployment [4]. PRESENCIFY utilizes an EdgeFace XS model checkpoint (gamma = 0.6) to generate 512-dimensional embeddings. Setyawan et al. developed FaceLiViT, combining linear vision transformers with structural reparameterization to reduce mobile inference latency [5]. Broader surveys highlight that edge benchmarks must report empirical latency alongside standard verification accuracy to provide realistic deployment guidance [6]. Taxonomic frameworks categorize facial recognition pipelines by template dimensionality, operational modality, and feature discriminability [7]. While scale-aware detectors such as YOLO-FaceV2 handle extreme occlusion effectively [8], PRESENCIFY adopts MediaPipe BlazeFace because its lightweight computational graph is better suited for dual-core and quad-core CPU execution.

B. Facial recognition attendance systems

Automated classroom attendance continues to see widespread experimental exploration in engineering literature [1]–[3]. Ahmad et al. assessed pretrained neural backbones embedded within a desktop attendance interface [1]. Patel et al. explored multi-student surveillance using fixed classroom CCTV infrastructure paired with convolutional networks [2]. Similarly,

Fan combined MobileNet feature extraction with sparrow search support vector machines to optimize classroom resource allocation [3]. Bibliometric surveys reflect expanding institutional interest in biometric recordkeeping [9]. Because existing experimental systems employ heterogeneous camera sensors, unstandardized room lighting, and private evaluation splits, direct numerical comparison across published results is methodologically flawed. PRESENCIFY focuses on transparent threshold boundaries, auditable review queues, and unvarnished error accounting.

Real-world deployments monitor individual students across complete lecture sessions rather than processing isolated static frames. Robust edge kiosks must suppress redundant consecutive captures, maintain immutable audit logs, and avoid recording transient unverified detections as absences. Ambient lighting fluctuations, head pose variations, and subtle facial expression changes alter cosine similarity scores dynamically. Evaluating a dual-threshold decision policy therefore requires examining intercepted borderline cases, administrative inspection burdens, and unmitigated high-confidence false acceptances simultaneously.

Successive video frames captured within milliseconds exhibit high visual redundancy and cannot serve as statistically independent trials. PRESENCIFY handles this operational constraint at runtime by binding attendance events directly to active session intervals and verified student identifiers. While these engineering safeguards streamline recordkeeping, they cannot repair degraded feature embeddings or substitute for physical spoof detection.

C. Presentation attacks, privacy and fairness

Face verification must be strictly distinguished from presentation attack detection (PAD) and anti-spoofing countermeasures [10]–[13]. Comprehensive literature reviews indicate that mitigating presentation attacks demands specialized optical sensors, dedicated classifier heads, and rigorous validation across diverse attack instruments [10]–[12]. Recent methodologies explore compact spatiotemporal architectures [14] and cross-domain video generalization [15]. PRESENCIFY incorporates no presentation attack detection modules and claims no resilience against printed photographs or screen replay attacks. Biometric privacy surveys confirm that unencrypted face templates remain vulnerable to model inversion and reconstruction attacks [16], [17]. Operating offline enhances data containment but does not relieve regulatory compliance obligations. Furthermore, demographic fairness analyses demonstrate that small convenience cohorts cannot validate equitable performance across diverse user populations [18].

III. METHODOLOGY

A. System architecture

Fig. 1 depicts the end-to-end hardware-aware processing pipeline from video capture to SQLite persistence. OpenCV acquires webcam frames through native operating system multimedia backends. MediaPipe BlazeFace isolates facial bounding boxes and extracts six anatomical landmarks in 3.5 ms on CPU. Affine transformation maps the detected facial crop into a normalized 112 by 112 pixel matrix in 0.8 ms. EdgeFace XS processes the aligned crop in 15.3 ms, generating a unit-normalized 512-dimensional embedding vector. SQLite manages student enrollment templates, session metadata, attendance logs, and administrative audit events. A modular Streamlit interface coordinates enrollment workflows, real-time scanning, instructor review queues, and attendance exports. Student registration mandates affirmative consent, and the user interface permanently displays a PAD_UNAVAILABLE status banner. Table I summarizes core system modules alongside their operational caveats.

TABLE I
REPORTED IMPLEMENTATION COMPONENTS AND CAVEATS

Component	Implementation	Role / Caveat
Detector	MediaPipe BlazeFace, CPU	Locates faces in 3.5 ms; detection does not identify persons.
Recognizer	EdgeFace XS, $\gamma = 0.6$, 512-d	Generates 512-d embeddings; operates under CC BY-NC-SA 4.0 terms.
Decision policy	Accept ≥ 0.65; review ≥ 0.52	Below 0.52 indicates low confidence, not verified absence.
Temporal guard	Three consecutive frames	Filters transient camera flicker; not multi-person tracking.
Storage	SQLite	Persists enrollment templates, session logs and review events.
PAD	PAD_UNAVAILABLE	No defenses against physical printouts or screen replay attacks.

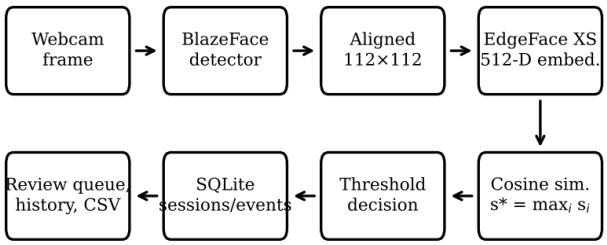


Fig. 1. PRESENCIFY implementation pipeline (CPU only). PAD is not part of the pipeline (PAD_UNAVAILABLE).

Attendance records follow session-scoped database upserts, preventing transient or sub-threshold detections from overwriting previously confirmed check-in timestamps. The administrative interface filters unresolved verification events by active lecture period, presenting instructors with candidate probe imagery for visual inspection. In physical lecture environments, commodity webcams regularly encounter variable illumination, motion blur, and unenrolled visitors. When the matcher assigns an UNKNOWN designation, this label indicates that matching confidence fell below operational cutoffs rather than confirming the student was absent from the room.

B. Scoring and decision policy

Let q denote a unit-normalized probe embedding and let g_i represent enrolled reference templates for active subjects. Equation (1) calculates pairwise cosine similarities and identifies the maximal match candidate index. Cosine similarity provides an uncalibrated relative score rather than a posterior probability of identity. Equation (2) defines the dual-threshold policy governed by an upper acceptance threshold tau_a = 0.65 and a lower review threshold tau_r = 0.52. Scores at or above tau_a trigger automated check-in; scores falling into the review band [tau_r, tau_a) divert to the instructor review queue; and scores below tau_r are rejected as UNKNOWN. At runtime, the

application requires three consecutive matching frames before committing attendance; this temporal guard was omitted from the static benchmark evaluation.

$$s_i = q^T g_i, \quad s^* = \max_{i} s_i \quad (1)$$

$$\text{Decision}(q) = \{ \text{ACCEPT: } s^* \geq 0.65; \text{ REVIEW: } 0.52 \leq s^* < 0.65; \text{ UNKNOWN: } s^* < 0.52 \} \quad (2)$$

C. Evaluation protocol and data

The empirical audit utilizes 410 facial captures partitioned into four distinct experimental splits, as summarized in Table II. The gallery split reserves 40 captures representing 20 enrolled subjects (two reference images per individual). The validation split comprises 100 captures designated for threshold exploration, strictly separated from subsequent testing. The primary held-out test split contains 166 static probe images, structured as 100 genuine probes from enrolled students and 66 impostor probes from unenrolled individuals. An unused split of 104 captures is set aside to maintain complete separation between experimental phases.

TABLE II
CAPTURE PARTITIONS IN THE CONTROLLED EVALUATION (410 CAPTURES)

Partition	Captures	Evaluation Purpose
Gallery	40	Enrolled reference templates (2 captures per subject across 20 students)
Validation	100	Threshold exploration split; completely separated from test probes
Held out test	166	Primary evaluation split: 100 genuine probes + 66 impostor probes
Unused	104	Excluded captures; prevents leakage across experimental stages
Total	410	Sum fully accounted for across all experimental partitions

Genuine test probes originate from enrolled subjects captured during separate imaging sessions, ensuring a capture-disjoint evaluation. Impostor probes are sourced exclusively from unenrolled individuals, guaranteeing an open-set identity-disjoint test protocol. Method A establishes a baseline single-threshold policy, accepting probes with similarity >= 0.65 and rejecting all lower scores as UNKNOWN. Method B introduces the uncertainty band, passing ambiguous probes between 0.52 and 0.65 to Stage 2 review. In the evaluation script, Stage 2 assigns outcomes using ground-truth subject metadata, establishing an idealized upper bound rather than measuring real human performance.

A critical methodological caveat concerns the provenance of probe images for subject P001. The evaluation pipeline attempted live webcam acquisition. When capture yielded fewer than ten facial crops, synthetic photometric variants were generated from a single capture; when acquisition failed entirely, the system substituted the initial ten images of the Olivetti research benchmark. These benchmark fallback images overlap with the initial slice used for subject P002. Because saved run logs do not definitively record which capture path executed, evaluators cannot verify that all 410 images represent authentic, independent webcam captures.

The repository contains test configuration discrepancies between documentation versions. Earlier summaries report 38 unit tests, whereas updated test executions confirm 53 passing test cases. These documentation variations underscore the importance of auditing executable test scripts directly rather than relying on static documentation badges.

D. Metrics

System performance is quantified using three primary metrics defined in Equations (3), (4), and (5). The False Acceptance Rate (FAR) measures the proportion of unauthorized impostor probes accepted as enrolled attendees. The Identity False Rejection Rate (FRR_id) quantifies genuine student probes that are either outright rejected or misidentified as another enrolled peer. The Review Rate (R_review) calculates the proportion of total probe traffic directed into the teacher review queue. These formulations strictly distinguish binary attendance decisions from fine-grained identity verification.

$$FAR = (\text{impostor probes accepted as enrolled}) / N_{imp} \quad (3)$$

$$FRR_{id} = (N_{gen\ rejected} + N_{gen\ misidentified}) / N_{gen} \quad (4)$$

$$R_{review} = N_{review} / N_{total} \quad (5)$$

IV. RESULTS AND DISCUSSION

A. Stored results

Table III summarizes the stored experimental results across the 166 held-out test probes (100 genuine, 66 impostor). Fig. 2 visually illustrates genuine probe outcomes and persistent impostor leakage across evaluation stages.

TABLE III
STORED TEST RESULTS (166 PROBES: 100 GENUINE, 66 IMPOSTOR).
STAGE 2 IS ORACLE SIMULATED, NOT OBSERVED HUMAN PERFORMANCE.

Measure	A: direct	B: Stage 1	B: Stage 2†	Observed Impact
Correct identity accepts	65 / 100	65 / 100	77 / 100	+12 genuine rescued
Misidentifications	7 / 100	7 / 100	7 / 100	Remains isolated
Genuine rej. / queued	28 rejected	5 rej. + 23 q.	16 rejected	Cut from 35% to 16%
Impostor false accepts	18 / 66	18 / 66	18 / 66	Upper tail (≥ 0.65)
Impostor rejections	48 / 66	15 rej. + 33 q.	48 / 66	Maintained across stages
Review workload	0 / 166 (0%)	56 / 166 (33.7%)	56 audited	Quantified human burden
Identity Precision	0.7222	None	0.7549	+3.27% improvement
Identity Recall	0.6500	None	0.7700	+12.00% improvement
Identity F1 Score	0.6842	None	0.7624	+0.0782 gain
Binary F1 Score	0.7579	None	0.8317	+0.0738 gain
Overall Accuracy	68.07%	None	75.30%	+7.23% net gain

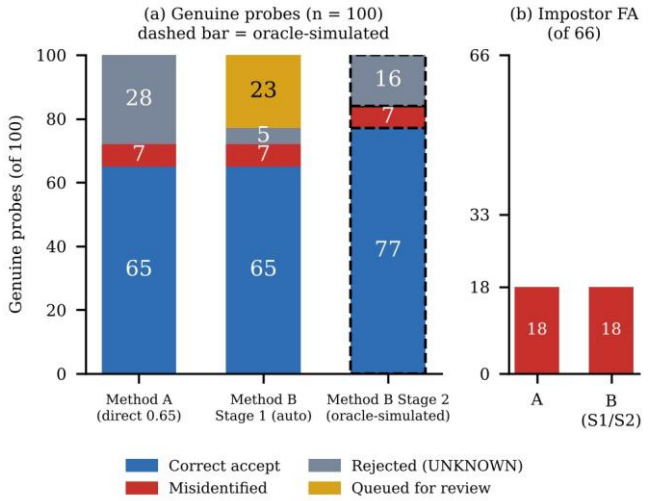


Fig. 2. Outcomes from the stored results. (a) Genuine probes (n = 100) per policy; the dashed outline Stage 2 bar is oracle simulated. (b) Impostor false accepts (n = 66) at s* ≥ 0.65, unchanged by review.

B. Automated stage

Under Method A, direct thresholding at 0.65 yields 65 correct genuine acceptances, 7 misidentifications, and 28 false rejections (FRR_id = 35.00%). Crucially, 18 of 66 impostor probes equal or exceed 0.65, resulting in an alarming 27.27% False Acceptance Rate. Because these high-scoring impostor probes match or exceed the automated acceptance cutoff, they bypass automated screening entirely. The review band cannot intercept them, demonstrating that static thresholding cannot eliminate confident proxy attendance attempts.

C. Review band and workload

The uncertainty review band [0.52, 0.65) intercepts 56 of the 166 test probes, generating a substantial 33.73% review burden. Queued items include 23 genuine probes and 33 borderline impostor probes (Fig. 3). Stage 1 automatically accepts 90 probes and rejects 20 probes without human intervention. While the review queue successfully intercepts 33 borderline impostor attempts, it requires instructors to manually inspect more than one-third of all attendance interactions.

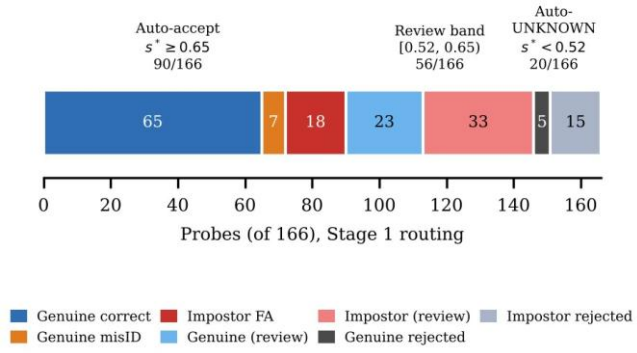


Fig. 3. Stage 1 routing of the 166 test probes under Method B, from stored counts: 90 probes are automatically accepted (65 correct, 7 misidentified genuine, 18 impostor false accepts), 56 are queued for review (23 genuine and 33 impostor) and 20 are automatically rejected (5 genuine, 15 impostor). Stage 2 is not shown because it is oracle simulated.

D. Simulated oracle review in Stage 2

Under oracle-simulated review, automated checking against ground-truth subject metadata recovers 12 genuine student probes and correctly rejects all 33 borderline impostor probes. Correct genuine verifications increase from 65 to 77, reducing false rejections to 16. This simulated recovery raises identity F1 from 0.6842 to 0.7624 and overall accuracy to 75.30%. Evaluators must recognize that these gains represent a theoretical mathematical upper bound rather than observed human instructor grading performance.

E. Consistency of the stored metrics

The stored summary metrics remain fully consistent with underlying probe frequency tallies. For Method A, 65 true positive accepts alongside 25 false positives produce an identity precision of 0.7222. Following simulated review, 77 true accepts alongside 25 false positives yield an identity precision of 0.7549. Binary attendance F1 reaches 0.7579 for Method A and climbs to 0.8317 under simulated review. Binary F1 scores should not be compared directly with identity F1 metrics because their operational definitions of positive cases differ.

F. Computational timing

Table IV details execution benchmarks gathered on an Intel Core i3-1115G4 processor without GPU acceleration. MediaPipe BlazeFace completes face detection in 3.5 ms, and affine alignment requires approximately 0.8 ms. EdgeFace XS extracts facial embeddings in 15.3 ms on average, yielding an aggregate computational latency of 19.6 ms per frame (approximately 51 FPS compute throughput). However, an earlier project document reports an average latency of 37 ms with peaks of 68 ms. Neither benchmark accounts for camera IO, Streamlit browser rendering, or multi-frame temporal buffering.

TABLE IV
REPORTED COMPUTATIONAL TIMINGS (INTEL CORE I3-1115G4, CPU ONLY).
COMPONENT TIMINGS, NOT END TO END WEBCAM THROUGHPUT.

Hardware Subsystem	Reported Value	Operational Caveat
EdgeFace XS pass	15.3 ms (mean)	Forward pass only; excludes frame capture
BlazeFace detector	3.5 ms (mean)	Detects face bounding box and 6 key landmarks
Affine alignment	0.8 ms (mean)	Warps cropped face to 112 × 112 canonical format
Aggregate execution	19.6 ms (≈51 FPS)	Total algorithm computation; excludes camera IO
Alternative report	37 ms avg., 68 ms max.	Unreconciled discrepancy found in older logs

G. Discussion

The dual decision thresholds partition separate regions of the similarity distribution. The upper cutoff tau_a distinguishes immediate passes from manual review candidates, while tau_r eliminates obvious non-matching faces. Because 18 impostors surpass 0.65, adjusting the review band cannot mitigate high-confidence proxy attendance leakage. The review queue functions as an operational triage buffer for borderline ambiguities rather than a universal security filter.

Borderline review queues contain more impostor probes (33) than genuine student captures (23). Consequently, instructors

allocate the majority of their review effort to rejecting unauthorized individuals rather than recovering legitimate students. Any empirical accuracy recovery for genuine students depends entirely on instructors making flawless visual verification judgments.

Existing attendance literature reports varied verification accuracies across bespoke hardware configurations and closed datasets [1]–[3]. Direct numerical score competition with those systems is methodologically invalid. PRESENCIFY instead contributes an open-set empirical audit documenting impostor leakage and human review workloads. Because standard metric matchers lack presentation attack detection [11], [13], high embedding similarity provides no guarantee against physical photograph spoofing.

H. Reproducibility and test status

Stored artifacts confirm that the four-stage evaluation script executed to completion with exit code 0. The automated test suite executes 53 passing test cases in 22.72 seconds following codebase refactoring. Earlier documentation records citing 38 tests reflect preliminary project stages. Git commit hashes b44205f and 0690d0a document verified codebase milestones. Future replications should execute the complete pipeline to verify individual probe scores directly.

I. What the evidence supports

Table V distinguishes claims directly supported by project artifacts from assumptions that lack empirical evidence. The prototype confirms feasible CPU-only inference but cannot guarantee flawless verification or physical spoof resilience.

TABLE V
SCOPE OF THE EVIDENCE

Supported Cautiously	Not Supported by Evidence
Working prototype workflow for registration, webcam scanning, attendance events and history export.	Production deployment across enterprise campus networks or unmonitored gates.
Dual threshold recognition with human review executed smoothly on commodity CPU hardware.	Guaranteed recognition accuracy or complete elimination of proxy attendance.
Manual review of borderline cases can rescue genuine false absences under ideal conditions.	Empirical validation of real world teacher review speed or human accuracy.
Transparent user interface disclosure of PAD_UNAVAILABLE status across all screens.	Presentation attack resistance, 3D mask spoof defense or photo replay detection.

J. Limitations

Multiple methodological constraints qualify the empirical scope of this study. Foremost among these is the complete absence of presentation attack detection (PAD_UNAVAILABLE), leaving the system vulnerable to printed photographs or screen replay attacks. Furthermore, 18 of 66 impostors achieved high-confidence automated acceptance, indicating that static thresholds cannot prevent determined proxy attendance. The Stage 2 review evaluation reflects an oracle simulation using ground-truth metadata rather than empirical classroom trials with practicing teachers. Additionally, the small 41-subject cohort lacks demographic diversity, challenging illumination, and wide pose variations. The capture provenance for subject P001 remains unresolved across stored logs. Finally, the three-frame temporal guard was omitted from static probe testing, reported latency benchmarks exhibit minor documentation variance, and facial biometric templates demand stringent legal safeguards, affirmative user consent, and transparent data deletion procedures [16], [17].

K. Threats to validity

Internal validity threats involve omitting temporal consistency checks during static testing and relying on ground-truth metadata during simulated reviews. If subject P001 utilized benchmark fallback images, genuine and impostor partitions share overlapping image slices. External validity is constrained because a small static evaluation cohort cannot capture the operational turbulence of crowded campus lecture halls. Reporting discrepancies in benchmark latencies and test counts require ongoing documentation synchronization. Moreover, the evaluation omitted formal confidence intervals and hypothesis testing.

Construct validity faces limitations because the investigation audits only a single pair of fixed decision thresholds. Raw cosine similarity distributions shift across different camera hardware, optical lens distortion, and ambient room lighting. Thresholds calibrated on one specific workstation cannot be assumed to function reliably in another environment. In the absence of human observer trials, simulated reviews cannot model real instructor fatigue, visual bias, or verification mistakes.

Threshold transportability presents an ongoing operational hurdle for edge biometric systems deployed across educational campuses. A similarity cutoff that separates identities under controlled lighting will drift when room illumination shifts. Until systems are evaluated across diverse physical classrooms, static benchmark numbers remain provisional engineering observations.

L. Privacy and responsible use

Facial embeddings constitute sensitive biometric data subject to rigorous legal and ethical governance [16], [17]. PRESENCIFY mandates affirmative student consent during registration and never persists raw video frames to disk. Production deployments must implement pseudonymous identifiers, database encryption, and automated record deletion policies. The EdgeFace XS model checkpoint carries a noncommercial research license (CC BY-NC-SA 4.0). Academic institutions must respect these licensing terms before sharing code or deploying systems commercially.

V. CONCLUSION

PRESENCIFY demonstrates that functional facial attendance workflows can execute locally on commodity laptop processors without dedicated graphics cards. Combining BlazeFace and EdgeFace XS achieves a 19.6 ms computational budget on CPU hardware. However, static decision thresholds permit 27.27% of unauthorized impostors to pass automated verification. Introducing an uncertainty review band intercepts borderline impostors but imposes a manual inspection workload on 33.73% of all attendance trials.

Simulated oracle review recovers genuine students only under the unrealistic assumption that instructors make zero misidentifications. The prototype transparently discloses its PAD_UNAVAILABLE status across all interfaces to prevent unjustified reliance on spoof resistance. Future research must evaluate active presentation attack detection modules, assemble

diverse classroom datasets, and conduct formal human-in-the-loop teacher verification trials.

As a college mini-project, PRESENCIFY succeeds by delivering a reproducible, inspectable edge biometric prototype. Rather than making unsubstantiated claims of high accuracy or security, the study transparently documents real edge bottlenecks, algorithmic trade-offs, and essential operational caveats.

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